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import os
import shutil
import re
import sys

# — TASK 3: Task Automation with Python
Scripts


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# Key concepts: os, shutil, re, requests,
file handling

# — Subtask A: Move all .jpg files from
one folder to another 

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def
move_jpg_files(source_folder="source_images
", dest_folder="moved_images"):
    """Move all .jpg files from
source_folder to dest_folder."""
    print("\n--- Subtask A: Move .jpg Files
---")

    # Create demo folders/files if they
don't exist
    if not os.path.exists(source_folder):
        os.makedirs(source_folder)
        # Create dummy jpg files for
demonstration
        for i in range(1, 4):
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open(os.path.join(source_folder,
f"photo{i}.jpg"), "w").close()

open(os.path.join(source_folder,
f"document{i}.txt"), "w").close()
    print(f"  Created demo folder
'{source_folder}' with sample files.")

    if not os.path.exists(dest_folder):
        os.makedirs(dest_folder)

    jpg_files = [f for f in
os.listdir(source_folder) if
f.lower().endswith(".jpg")]

    if not jpg_files:
        print(f"  No .jpg files found in
'{source_folder}'.")
        return

    moved = 0
    for filename in jpg_files:
        src_path =
os.path.join(source_folder, filename)
        dst_path =
os.path.join(dest_folder, filename)
        shutil.move(src_path, dst_path)
        print(f"  ✓ Moved: {filename} →
{dest_folder}/")
        moved += 1

    print(f"  Done! {moved} file(s) moved
to '{dest_folder}'.")

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# — Subtask B: Extract all email
addresses from a .txt file


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def
extract_emails(input_file="emails_input.txt
", output_file="emails_found.txt"):
    """Extract all email addresses from a
.txt file and save them."""
    print("\n--- Subtask B: Extract Emails
from Text File ---")

    # Create a demo input file if not
present
    if not os.path.exists(input_file):
        demo_content = """
Hello team,

Please contact support@codealpha.tech or
services@codealpha.com for help.
You can also reach admin@company.org or
hr.dept@workplace.in.
Invalid: not-an-email, @missinguser.com,
missingdomain@

Thanks,
The Team <noreply@newsletter.io>
"""

        with open(input_file, "w") as f:
            f.write(demo_content)
        print(f" Created demo file
'{input_file}'.")

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with open(input_file, "r") as f:
    content = f.read()

    email_pattern = r"[a-zA-Z0-9._%+\-]+@[a-zA-Z0-9.\-]+\.[a-zA-Z]{2,}"
    emails = re.findall(email_pattern,
content)
    unique_emails =
list(dict.fromkeys(emails)) # preserve
order, remove dups

    if not unique_emails:
        print(" No emails found.")
        return

    with open(output_file, "w") as f:
        f.write("Extracted Email
Addresses\n")
        f.write("=" * 30 + "\n")
        for email in unique_emails:
            f.write(email + "\n")

    print(f" Found {len(unique_emails)}
unique email(s):")
    for email in unique_emails:
        print(f"      • {email}")
    print(f" ✓ Saved to
'{output_file}'.")

# — Subtask C: Scrape the title of a
fixed webpage and save it

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def
scrape_webpage_title(url="https://www.examp
le.com", output_file="page_title.txt"):
    """Fetch a webpage and extract its
    <title> tag, then save to file."""
    print("\n--- Subtask C: Scrape Webpage
    Title ---")

    try:
        import urllib.request
        headers = {"User-Agent":
"Mozilla/5.0"}
        req = urllib.request.Request(url,
headers=headers)
        with urllib.request.urlopen(req,
timeout=10) as response:
            html =
response.read().decode("utf-8",
errors="ignore")

            title_match = re.search(r"
<title[^>]*>(.*?)</title>", html,
re.IGNORECASE | re.DOTALL)
            title =
title_match.group(1).strip() if title_match
else "No title found"

        print(f" URL : {url}")
        print(f" Title : {title}")

        with open(output_file, "w") as f:
            f.write(f"URL: {url}\n")
            f.write(f"Title: {title}\n")

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        print(f" ✓ Saved to
'{output_file}'.")

    except Exception as e:
        print(f" ⚠ Could not fetch page:
{e}")

        # Save the error gracefully
        with open(output_file, "w") as f:
            f.write(f"URL: {url}\nError:
{e}\n")

# — Main Menu



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def run_automation():
    print("\n" + "=" * 50)
    print(" 🤖 Task Automation with
Python Scripts")
    print("=" * 50)
    print("\n Select a subtask to run:")
    print(" [1] Move .jpg files to a new
folder")
    print(" [2] Extract emails from a .txt
file")
    print(" [3] Scrape title of a
webpage")
    print(" [4] Run ALL subtasks")
    print(" [0] Exit")

    choice = input("\n Your choice:
").strip()

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        if choice == "1":
            move_jpg_files()
        elif choice == "2":
            extract_emails()
        elif choice == "3":
            url = input(" Enter URL (press
Enter for example.com): ").strip()
            scrape_webpage_title(url if url
else "https://www.example.com")
        elif choice == "4":
            move_jpg_files()
            extract_emails()
            url = input("\n Enter URL for
title scraping (Enter = example.com):
").strip()
            scrape_webpage_title(url if url
else "https://www.example.com")
        elif choice == "0":
            print("\n Exiting. Goodbye!\n")
            sys.exit()
        else:
            print(" ⚠ Invalid choice.")

        print("\n Automation complete!\n")

if __name__ == "__main__":
    run_automation()

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